

# Understanding Supply and Demand Disparities in EV Charging Infrastructure

## *A Variable Framework for Geographic and Behavioral Analysis*

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*This work was completed as an independent research contribution to a larger faculty-led project at UC San Diego. The overarching research agenda was led by a PhD researcher in the Economics Department. The variable framework, R-based data pipeline, and behavioral economics analysis documented here represent the author's independent contribution.*

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## Abstract

This paper investigates the geographic and socioeconomic factors driving disparities between the supply and demand of public electric vehicle (EV) charging infrastructure across California communities. Using a quantitative modeling framework built on zip-code-level data from the California Department of Motor Vehicles (DMV), U.S. Census Bureau, and infrastructure datasets, this research categorizes vehicles by manufacturer, fuel type, and charging plug compatibility to identify infrastructure gaps. Beyond documenting where gaps exist, this analysis examines why they exist, drawing on economic, behavioral, and environmental variables to explain local variation in both charging demand and network supply.

The research further connects empirical findings to behavioral economic theory, examining how rational inattention, price salience, and nudge theory shape consumer charging behavior and firm pricing strategy under varying levels of price transparency. This variable framework is intended to support more accurate demand forecasting models and to inform data-driven infrastructure investment decisions.

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## 1. Introduction

The rapid growth of electric vehicle adoption across the United States has created an urgent need to understand the spatial and behavioral dynamics that determine where public charging infrastructure is needed, and where it is being built. A consistent pattern emerges in the data: supply and demand for EV charging are frequently misaligned. In some areas, charging stations sit underutilized while nearby communities with high EV ownership face persistent shortages. In others, infrastructure has been deployed in locations that do not reflect actual demand patterns.

This mismatch reflects the complex interaction of economic conditions, housing characteristics, transportation behavior, land use patterns, local policy environments, and consumer psychology. Standard demand models that rely solely on EV ownership counts as a proxy for charging need are insufficient — they fail to capture the contextual factors that determine whether a household relies on public charging, when and where they charge, and how they respond to pricing signals.

This research constructs a comprehensive variable framework organized across five domains: socioeconomic and housing characteristics, transportation and commuting behavior, physical infrastructure, energy and environmental conditions, and policy context. A sixth domain addresses behavioral variables derived from consumer activity patterns.

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## 2. Research Question

The central question guiding this research is:

***Why do EV charging supply and demand differ across different geographic areas, and which community-level factors best explain this variation?***

EV ownership provides a direct but incomplete measure of potential charging demand. Community-level contextual factors — including housing type, income, commuting behavior, infrastructure access, and local policy — shape both the scale and the spatial distribution of that demand, as well as the conditions under which charging network operators choose to invest in supply.

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## 3. Data Sources

This analysis draws on the following primary data sources:

- **California DMV:** Raw vehicle registration records converted into zip-code-level datasets categorizing vehicles by manufacturer, fuel type, and charging plug type.
  - **U.S. Census Bureau, ACS:** Household income, housing tenure and type, commute patterns, educational attainment, employment sector, and demographic data at the zip code level.
  - **Alternative Fuels Station Locator (DOE/AFDC):** Public EV charging station locations, network operators, charger types, and installation dates.
  - **EPA Air Quality Index (AQI):** Regional air quality measurements used as a proxy for environmental policy pressure.
  - **U.S. Energy Information Administration (EIA):** Regional electricity prices by utility service area.
  - **OpenStreetMap and GIS data:** Proximity to highways, shopping centers, restaurants, and parking infrastructure.
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## 4. Variable Framework

The following sections define each variable category used in this analysis, along with the theoretical rationale connecting each variable to EV charging supply or demand.

### 4.1 Socioeconomic and Housing Variables

Housing type and tenure are among the strongest predictors of public charging reliance. A household with access to a private garage can charge at home overnight, dramatically reducing dependence on public infrastructure. Conversely, households in multi-unit dwellings or rental properties face structural barriers to home charging installation, pushing them toward public networks regardless of EV

ownership rates.

- **Average Household Income:** Higher incomes correlate with higher EV adoption rates, reflecting both purchase price and environmental awareness. However, high-income areas with single-family housing may generate less public charging demand due to home charging access.
- **Share of Single-Family Homes:** Areas with greater single-family housing density are associated with higher rates of home charging access, reducing per-capita public charging demand.
- **Share of Renter-Occupied Housing Units:** Renters face significant barriers to installing home charging equipment, making this a direct proxy for structural public charging demand.
- **Average Number of Rooms per Housing Unit:** Used as a secondary proxy for housing spaciousness and the likelihood of private parking or garage access.
- **Population Density:** Higher density areas have less access to private parking and charging, increasing reliance on public networks. Density also affects the economics of network deployment.
- **Median Age:** Younger populations have shown higher rates of early EV adoption. Age distribution correlates with technology adoption curves relevant to EV uptake.
- **Educational Attainment:** Higher educational attainment is associated with greater environmental awareness and earlier adoption of green technologies.
- **Employment Sector Composition:** Regions with higher concentrations of professional and technology workers tend to exhibit higher EV ownership rates, driven by both income and workplace sustainability culture.

## 4.2 Transportation and Commuting Variables

Commuting behavior determines when and how frequently a vehicle requires charging. Long-distance commuters deplete battery charge more rapidly and may require public charging during the workday. The growth of remote work has introduced new spatial patterns in charging demand, shifting activity from workplace corridors to residential and amenity-rich environments.

- **Transportation Mode to Work:** The share of workers commuting by private vehicle determines baseline daily charging demand.
- **Average Commute Distance or Time:** Longer commutes result in greater daily battery depletion, increasing the probability of mid-day or workplace charging needs.
- **Time of Departure for Work:** Departure time distributions provide insight into peak demand windows for workplace and corridor charging.
- **Share of Workers Who Work From Home:** Remote workers shift public charging demand away from workplace-adjacent stations toward daytime amenity locations such as shopping centers and recreational areas.

## 4.3 Physical Infrastructure Variables

The built environment directly shapes both the demand for public charging and the locations where operators choose to deploy stations. Parking availability affects home charging feasibility. Co-location with retail and dining establishments improves station utilization.

- **Parking Availability:** The share of households without a garage or driveway is a direct structural barrier to home charging and one of the strongest predictors of public charging demand.
- **Proximity to Highways and Major Roads:** High-traffic corridors generate demand for DC Fast Charging from both local EV owners and intercity travelers, and shape operators' site selection decisions.
- **Number of Retail, Dining, and Amenity Establishments Nearby:** EV charging stations are frequently co-located where drivers already spend time, improving expected utilization rates and the economic case for network investment.

#### 4.4 Energy and Environmental Variables

Electricity pricing affects both the cost of public charging and the relative attractiveness of home versus public charging. Environmental conditions serve as a proxy for regulatory and social pressure driving EV adoption policy.

- **Electricity Prices by Region:** Higher electricity costs reduce the cost advantage of EVs and may discourage frequent public charging among cost-sensitive consumers. Regional price variation also affects charging network profitability.
- **Air Quality Index (AQI):** Areas with chronically poor air quality are more likely to be subject to strong state and local EV adoption incentives, zero-emission vehicle mandates, and infrastructure subsidies.

#### 4.5 Policy and Incentive Variables

Policy interventions operate on both sides of the supply-demand equation. Consumer-facing incentives directly stimulate EV adoption and charging demand. Supply-side subsidies and mandates accelerate network deployment by reducing operator costs and regulatory barriers.

- **Local and State EV Purchase Incentives:** Rebates and tax credits reduce effective purchase price, accelerating adoption and expanding the income range of households likely to own EVs.
- **Clean Energy Mandates and Sustainability Commitments:** Jurisdictions with binding carbon reduction targets tend to prioritize EV infrastructure investment in both public sector procurement and land use regulation.
- **Charger Installation Subsidies and Grants:** Direct financial support for charging station installation reduces operator capital costs and lowers the utilization threshold required for profitability.

#### 4.6 Behavioral and Activity Variables

- **Tourism and Visitation Rates:** High-traffic destinations such as coastal areas, urban entertainment districts, and shopping centers generate substantial transient charging demand from EV-owning visitors.
- **Number of Registered Vehicles Per Capita:** Serves as a baseline for car ownership intensity and a direct proxy for EV adoption potential.

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## 5. Behavioral Economic Dimensions

The standard economic model of EV charging assumes consumers respond rationally to price signals, selecting stations based on cost, proximity, and availability. Behavioral economic theory suggests a more complex picture. Three mechanisms are particularly relevant to this analysis.

### **5.1 Rational Inattention**

Rational inattention theory proposes that consumers face cognitive costs in processing information. When the mental cost of evaluating price differences across charging stations exceeds the expected financial benefit of switching, consumers rationally ignore price signals. This has direct implications for the relationship between price transparency and station utilization — even where dynamic pricing data is publicly available, consumers may default to habitual station choices, reducing the effectiveness of price-based demand management strategies.

### **5.2 Price Salience and Nudging**

The salience of pricing information significantly affects consumer response. Interventions such as green visual indicators for low-price periods, roadside signage, and in-app notifications represent attempts to make price information more salient at the moment of decision. However, these nudges may have limited effectiveness for habitual behaviors, where the decision of where to charge is made automatically rather than deliberatively.

This analysis examines variation in consumer response to pricing signals across demographic groups, with the hypothesis that higher-income and higher-education populations are more likely to engage in deliberate station selection, while lower-income populations may show greater price responsiveness due to budget constraints.

### **5.3 Market Competition and Strategic Pricing**

Charging network operators adapt their pricing strategies in response to consumer behavioral patterns. Where consumers exhibit rational inattention or strong habit formation, operators face reduced competitive pressure to lower prices, even in markets with multiple providers. This dynamic can contribute to persistently high public charging costs in areas where behavioral factors reduce price sensitivity, potentially undermining the cost advantage of EV ownership for drivers without home charging access.

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## **6. Methodology Overview**

The empirical analysis proceeds in three stages. First, raw DMV vehicle registration data is processed using an R-based pipeline to generate zip-code-level metrics for EV ownership, vehicle type distribution, and plug type compatibility. This dataset is merged with Census ACS variables and infrastructure data to construct a comprehensive cross-sectional dataset at the zip code level.

Second, regression-based models are estimated to identify the independent contribution of each variable category to observed variation in public charging demand and supply. Spatial autocorrelation is addressed through the inclusion of geographic fixed effects and spatially lagged dependent variables.

Third, behavioral hypotheses regarding rational inattention and price responsiveness are tested using variation in local price transparency policy and pricing data from charging network operators.

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## 7. Expected Contributions

This research makes several contributions to the literature on EV infrastructure planning and behavioral economics. First, it provides a systematic variable framework integrating supply-side, demand-side, and behavioral dimensions into a single analytical structure, offering a more complete account of infrastructure misalignment than models focused solely on EV ownership counts.

Second, by connecting quantitative geographic analysis to behavioral economic theory, this research highlights the limitations of purely price-based approaches to demand management and suggests that infrastructure planning must account for the cognitive and behavioral factors that shape how consumers interact with public charging networks.

Third, the zip-code-level granularity of the analysis enables identification of specific community types that are systematically underserved by existing infrastructure, providing a basis for more targeted and equitable investment strategies.

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